Multi-model nowcasting and short-term forecasting of infectious diseases

Johannes Bracher

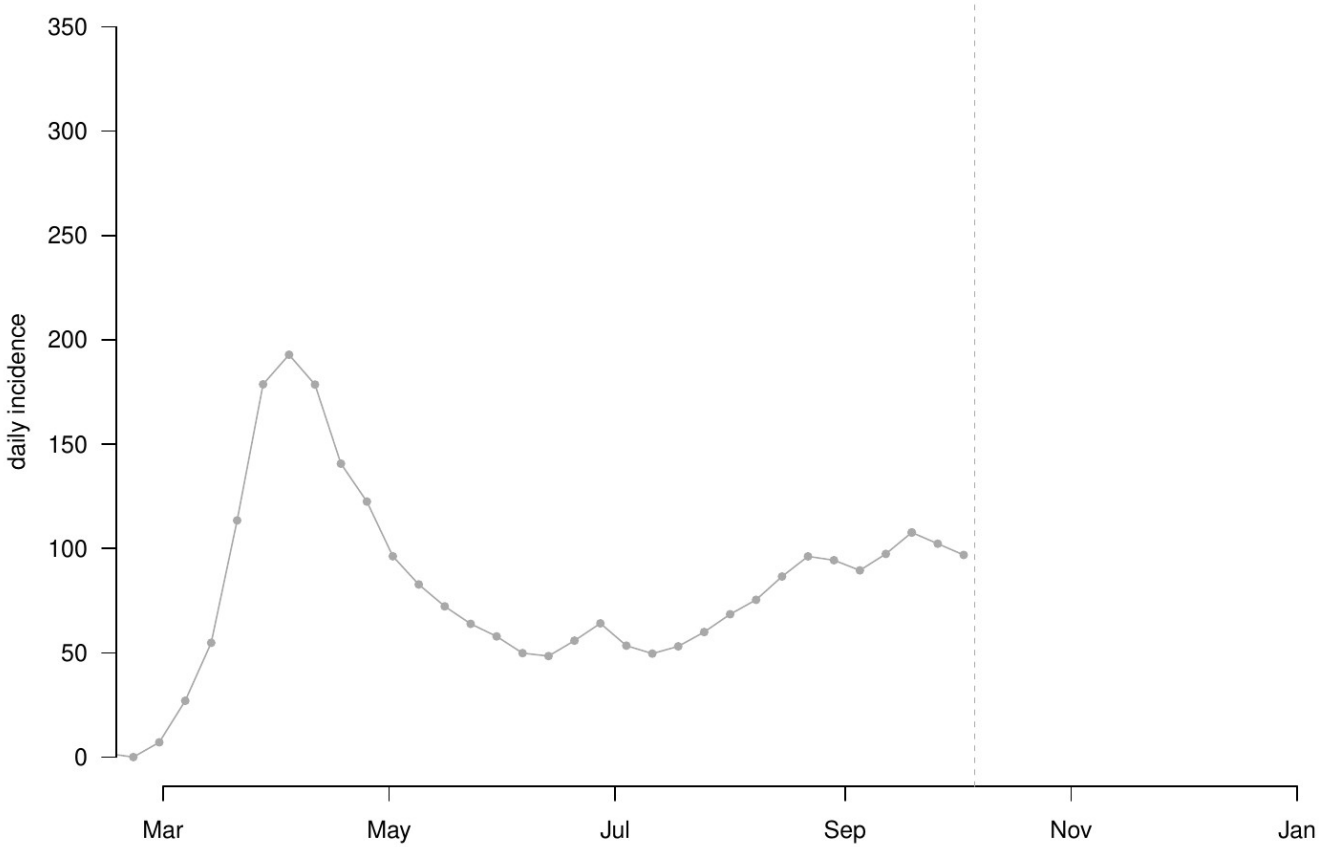
ISCB GMDS 2026, 28 Sep 2026



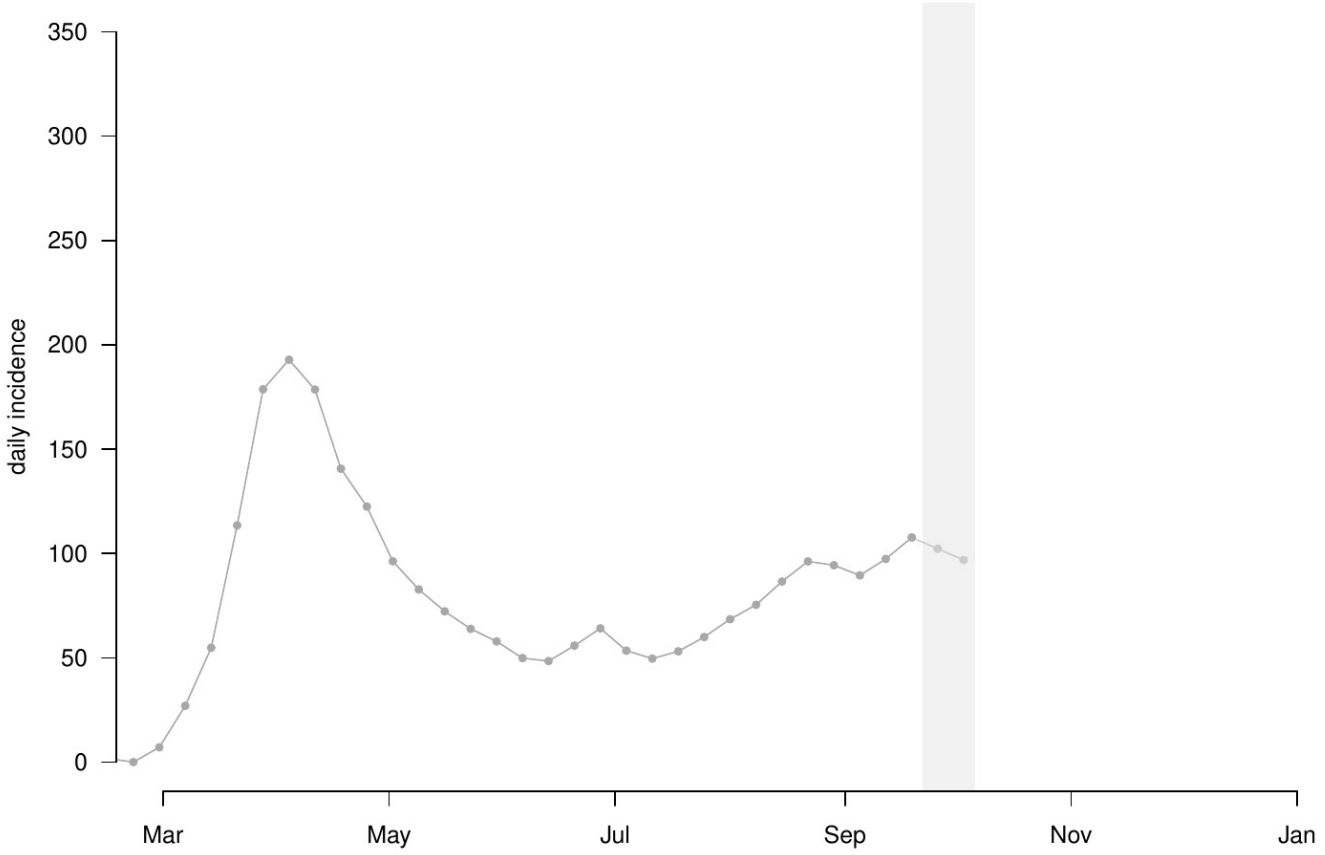


What is this all about?

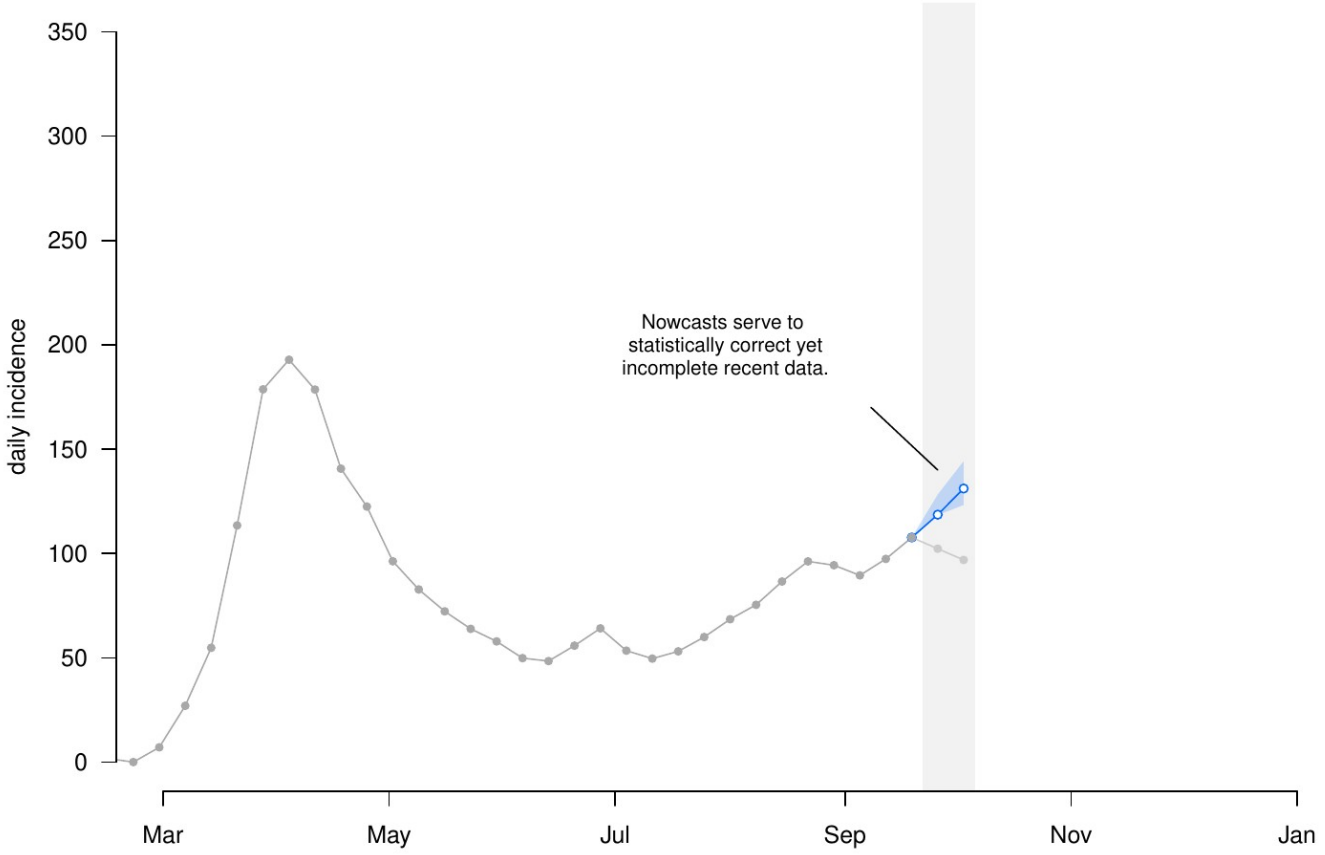
Predictive infectious disease modelling



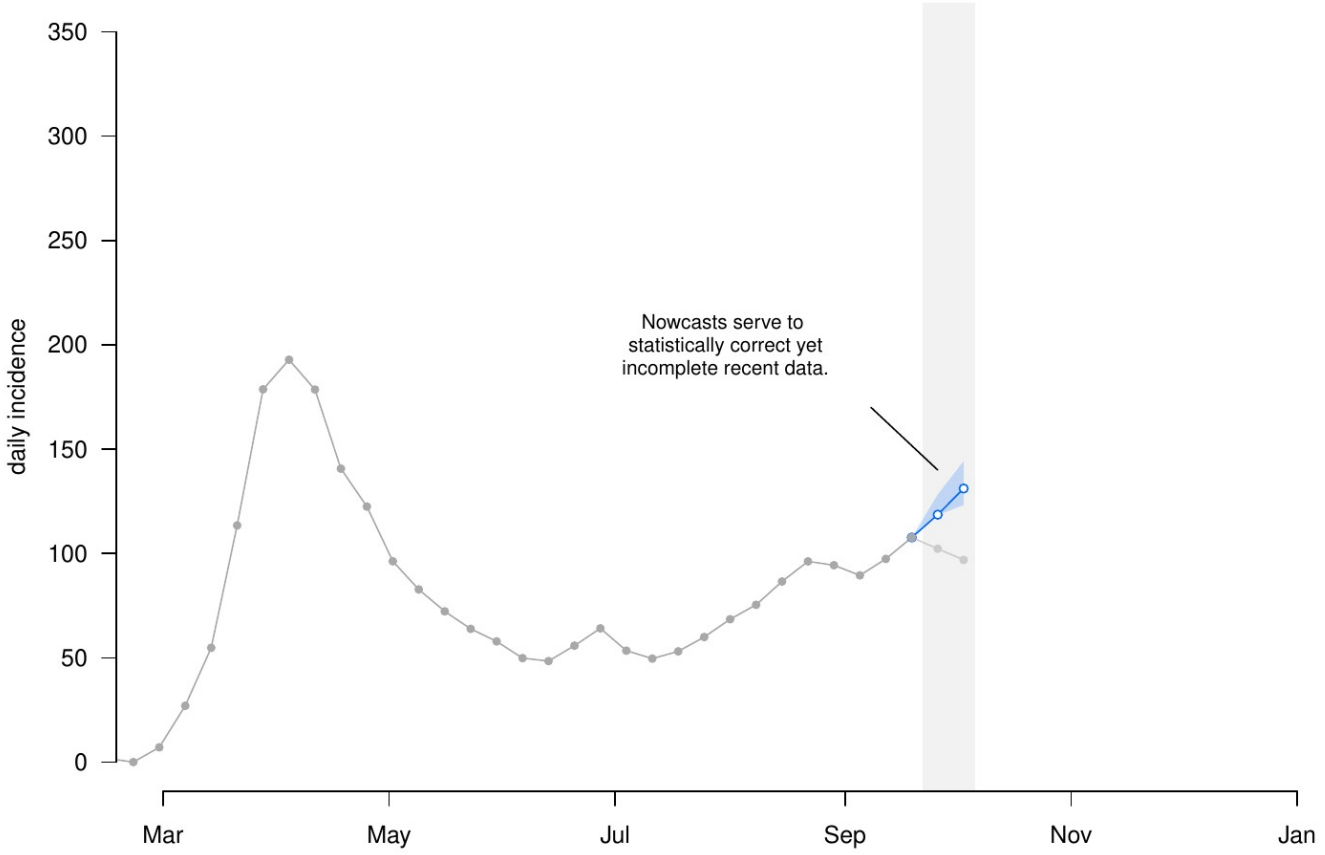
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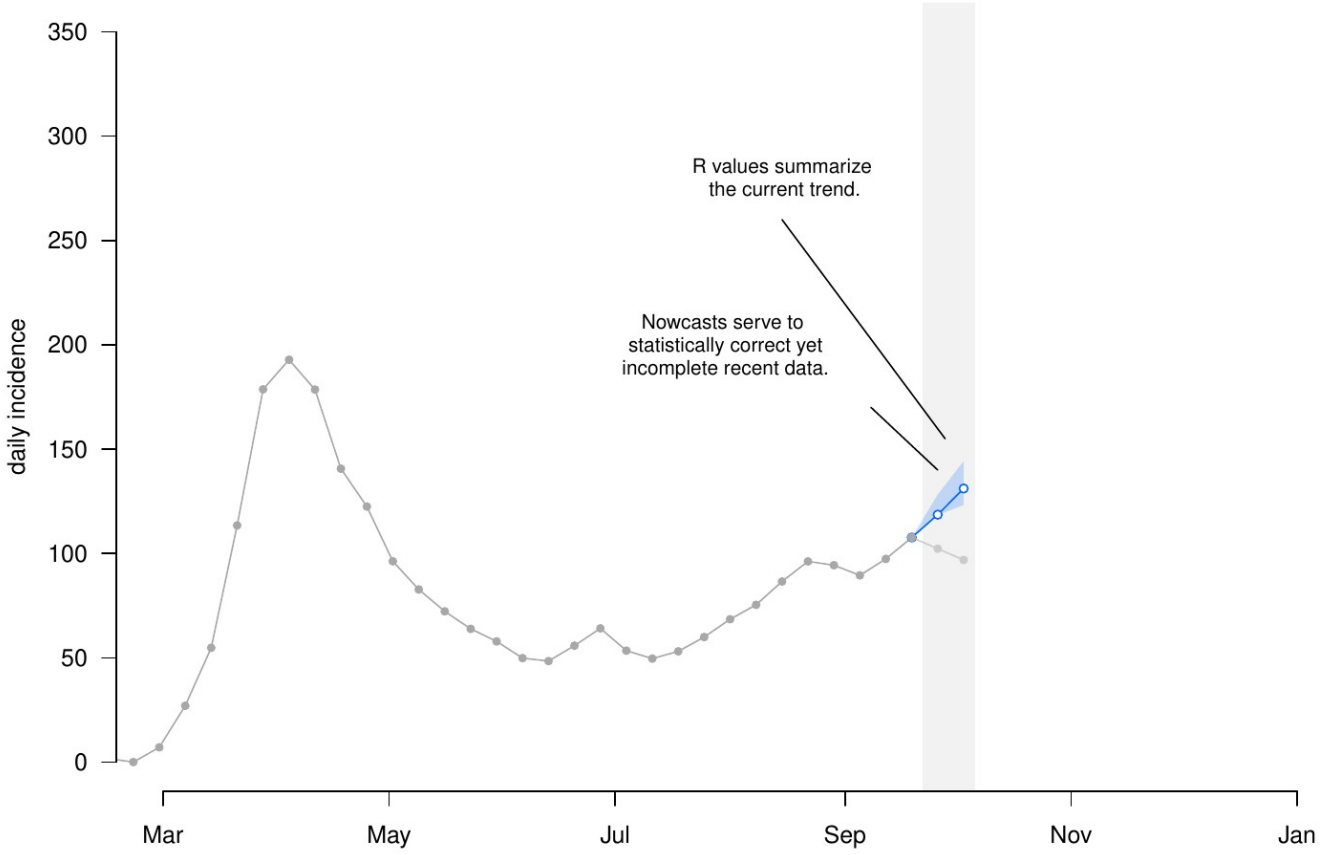
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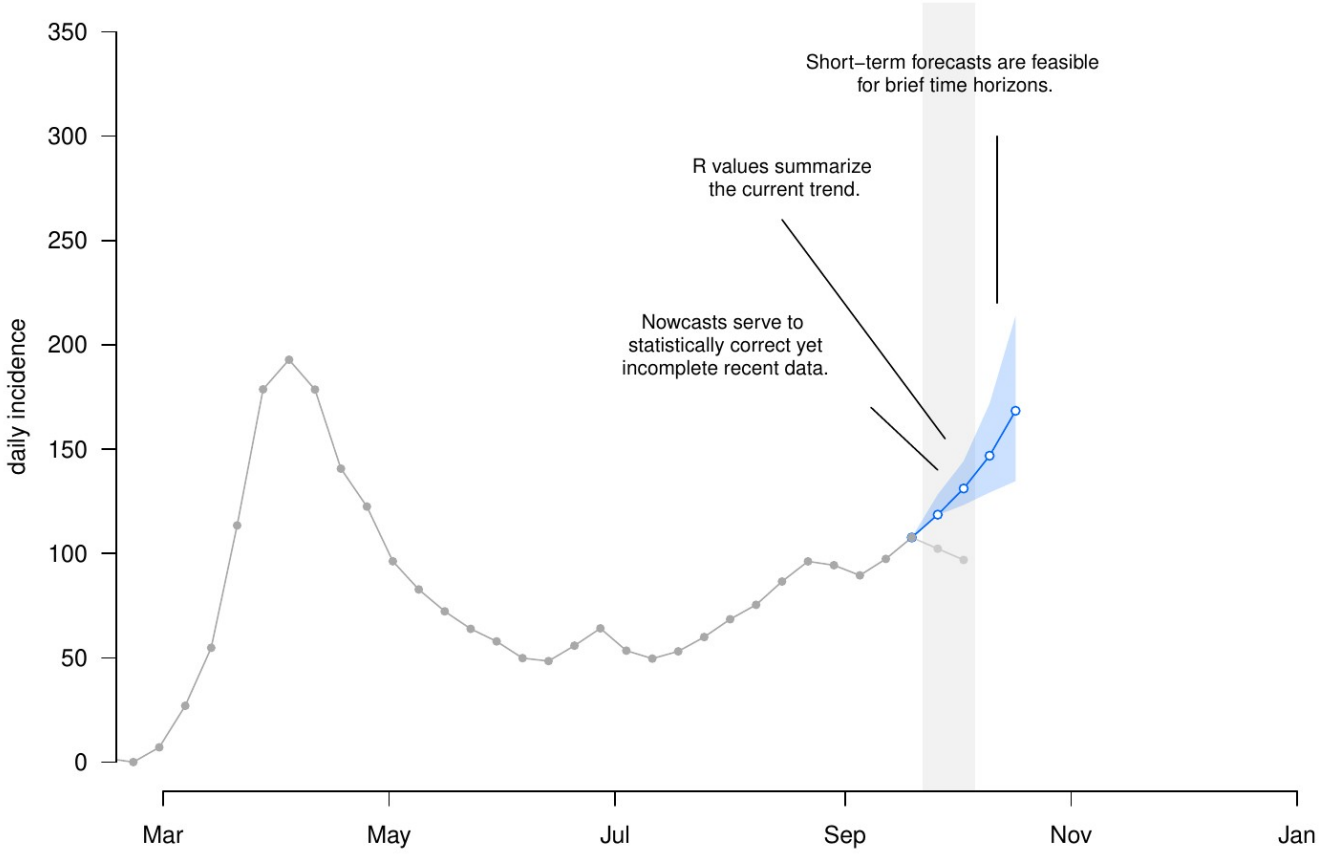
Predictive infectious disease modelling



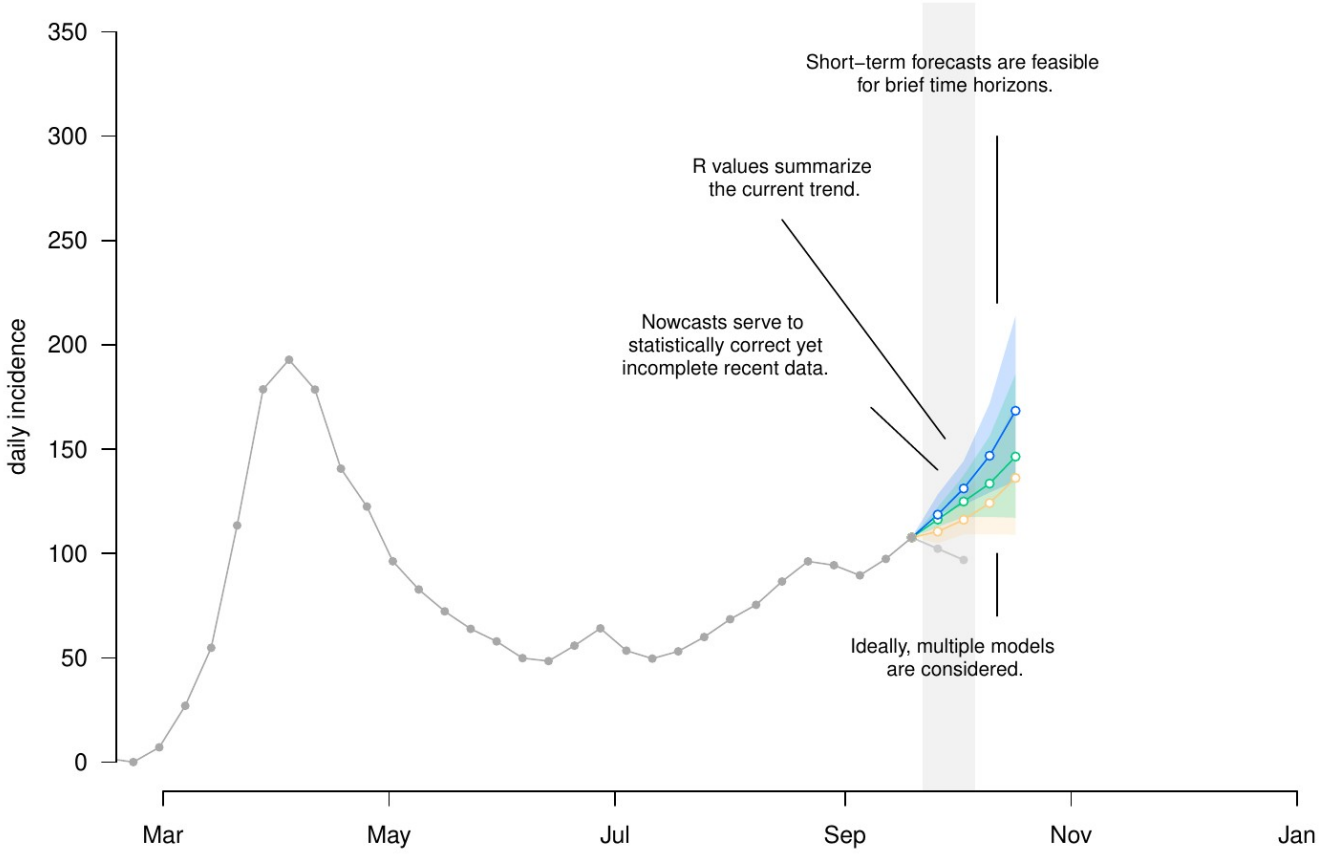
Predictive infectious disease modelling



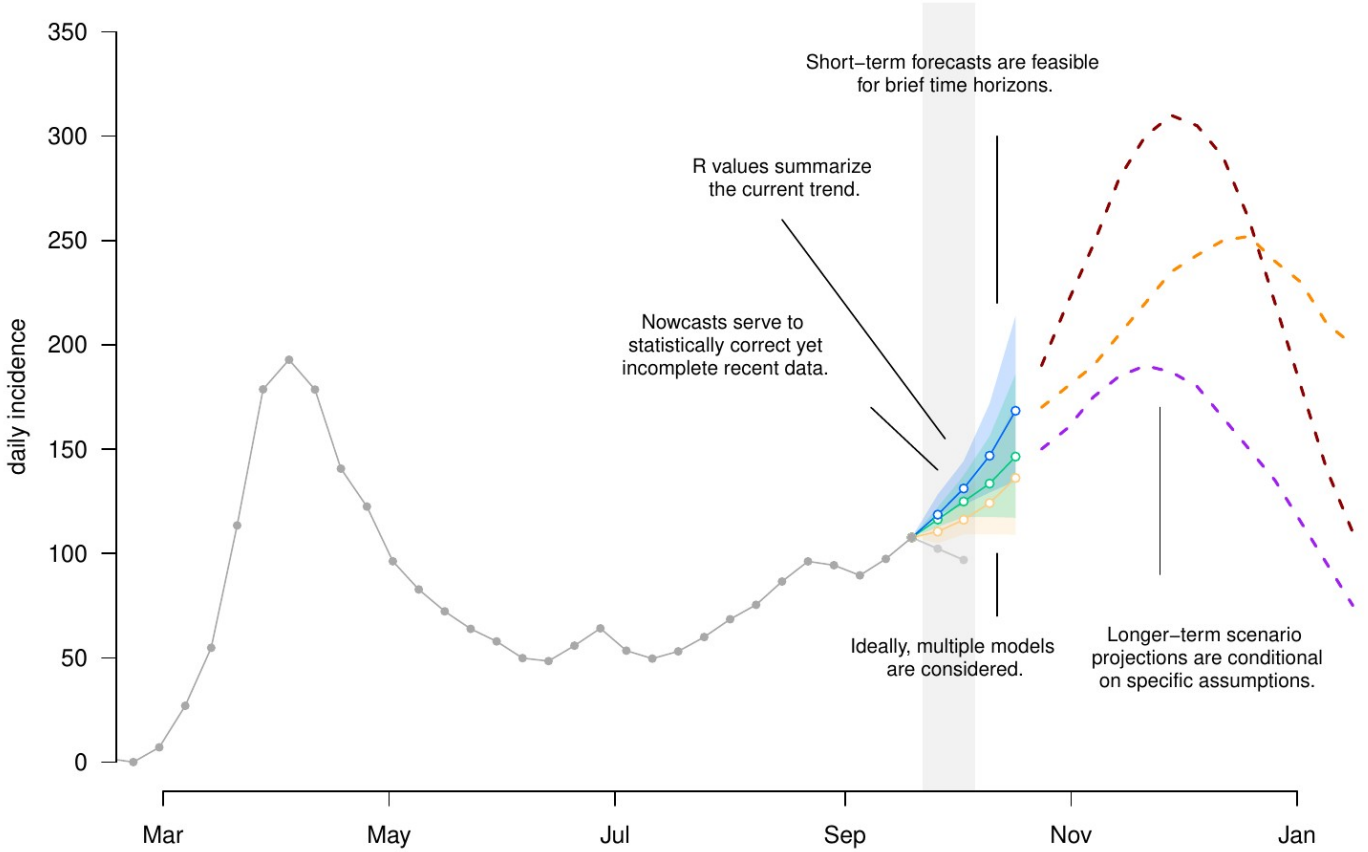
Predictive infectious disease modelling



Predictive infectious disease modelling

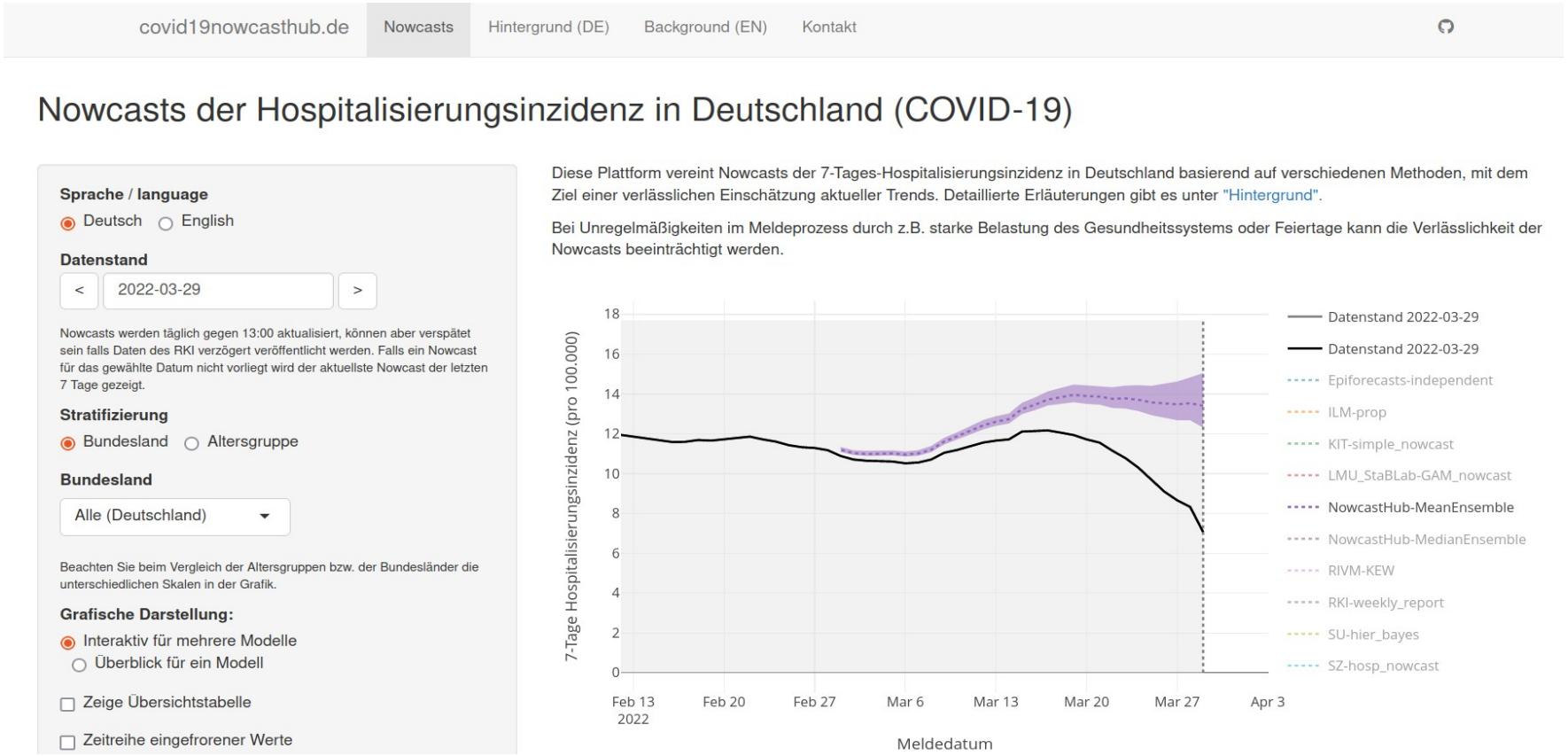


Predictive infectious disease modelling



Collaborative Hubs and ensemble predictions

- Example: nowcasting COVID-19 Hospitalizations in Germany



Collaborative Hubs and ensemble predictions

- Example: nowcasting COVID-19 Hospitalizations in Germany

covid19nowcasthub.de | Nowcasts | Hintergrund (DE) | Background (EN) | Kontakt

Nowcasts der Hospitalisierungsinzidenz in Deutschland (COVID-19)

Sprache / language

(•) Deutsch ( ) English

Datenstand

< 2022-03-29 >

Nowcasts werden täglich gegen 13:00 aktualisiert, können aber verspätet sein falls Daten des RKI verzögert veröffentlicht werden. Falls ein Nowcast für das gewählte Datum nicht vorliegt wird der aktuellste Nowcast der letzten 7 Tage gezeigt.

Stratifizierung

(•) Bundesland ( ) Altersgruppe

Bundesland

Alle (Deutschland)

Beachten Sie beim Vergleich der Altersgruppen bzw. der Bundesländer die unterschiedlichen Skalen in der Grafik.

Grafische Darstellung:

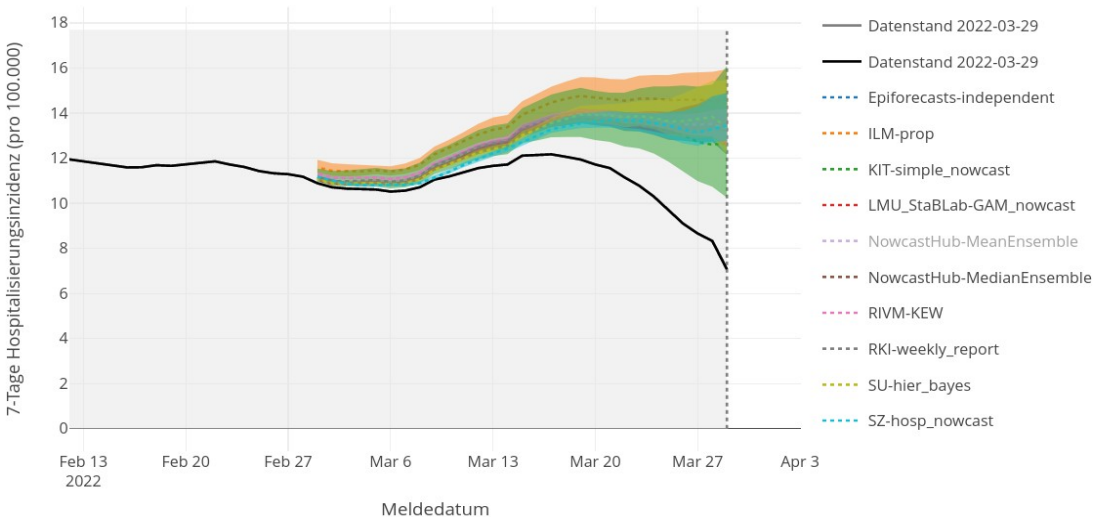
(•) Interaktiv für mehrere Modelle
( ) Überblick für ein Modell

[ ] Zeige Übersichtstabelle

[ ] Zeitreihe eingefrorener Werte

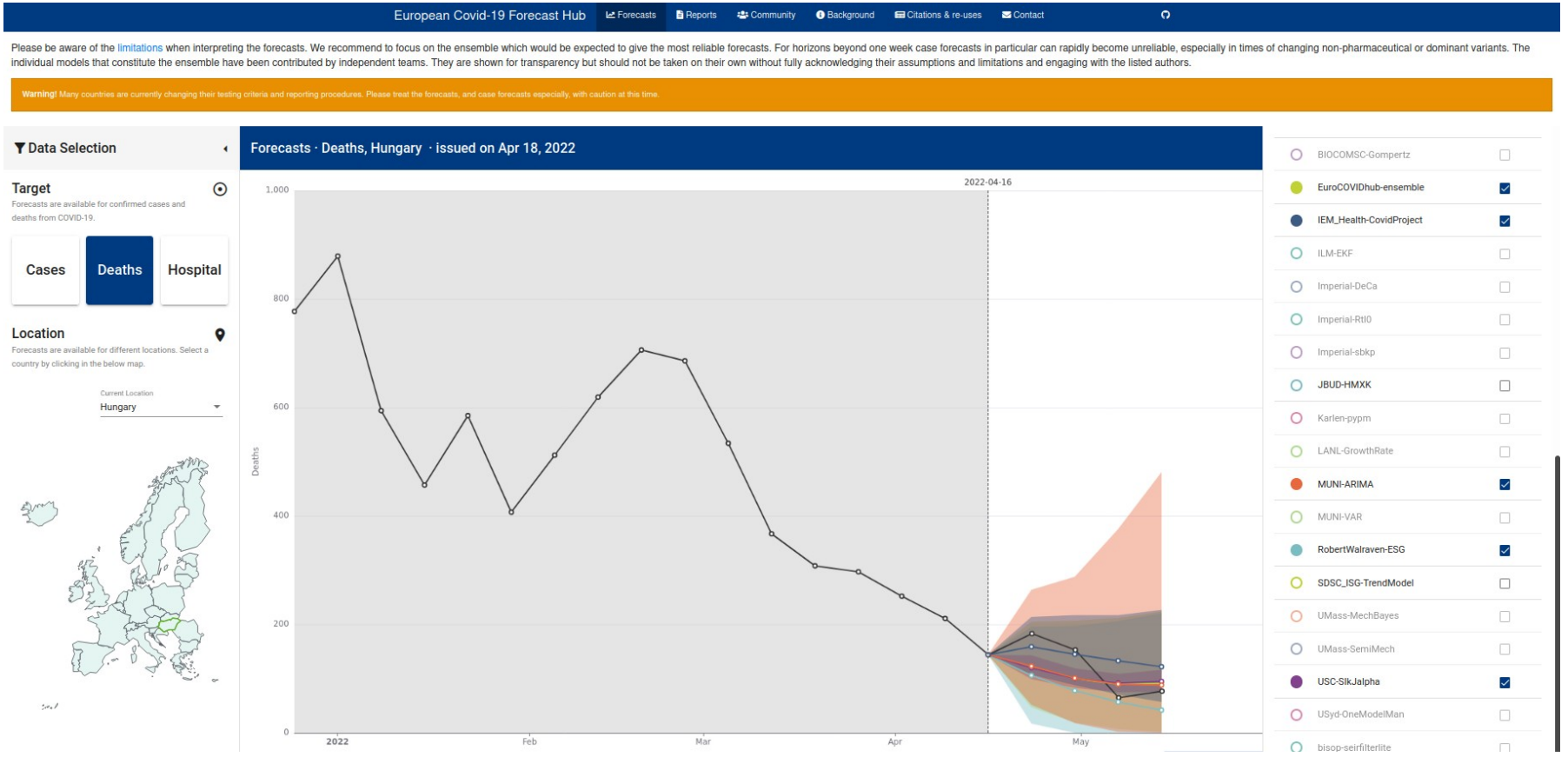
Diese Plattform vereint Nowcasts der 7-Tages-Hospitalisierungsinzidenz in Deutschland basierend auf verschiedenen Methoden, mit dem Ziel einer verlässlichen Einschätzung aktueller Trends. Detaillierte Erläuterungen gibt es unter "Hintergrund".

Bei Unregelmäßigkeiten im Meldeprozess durch z.B. starke Belastung des Gesundheitssystems oder Feiertage kann die Verlässlichkeit der Nowcasts beeinträchtigt werden.



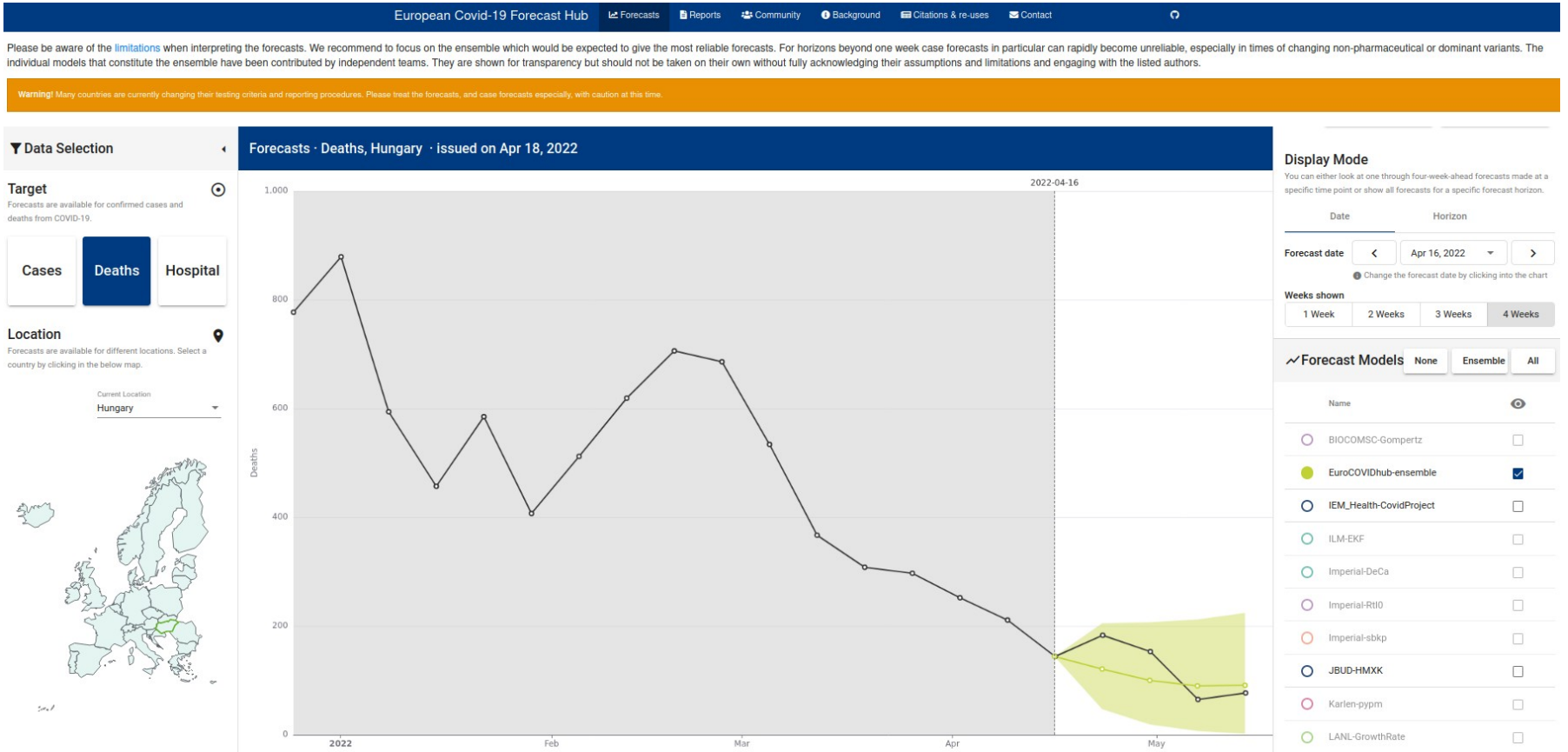
Collaborative Hubs and ensemble predictions

- Example: ECDC Forecasting Hub



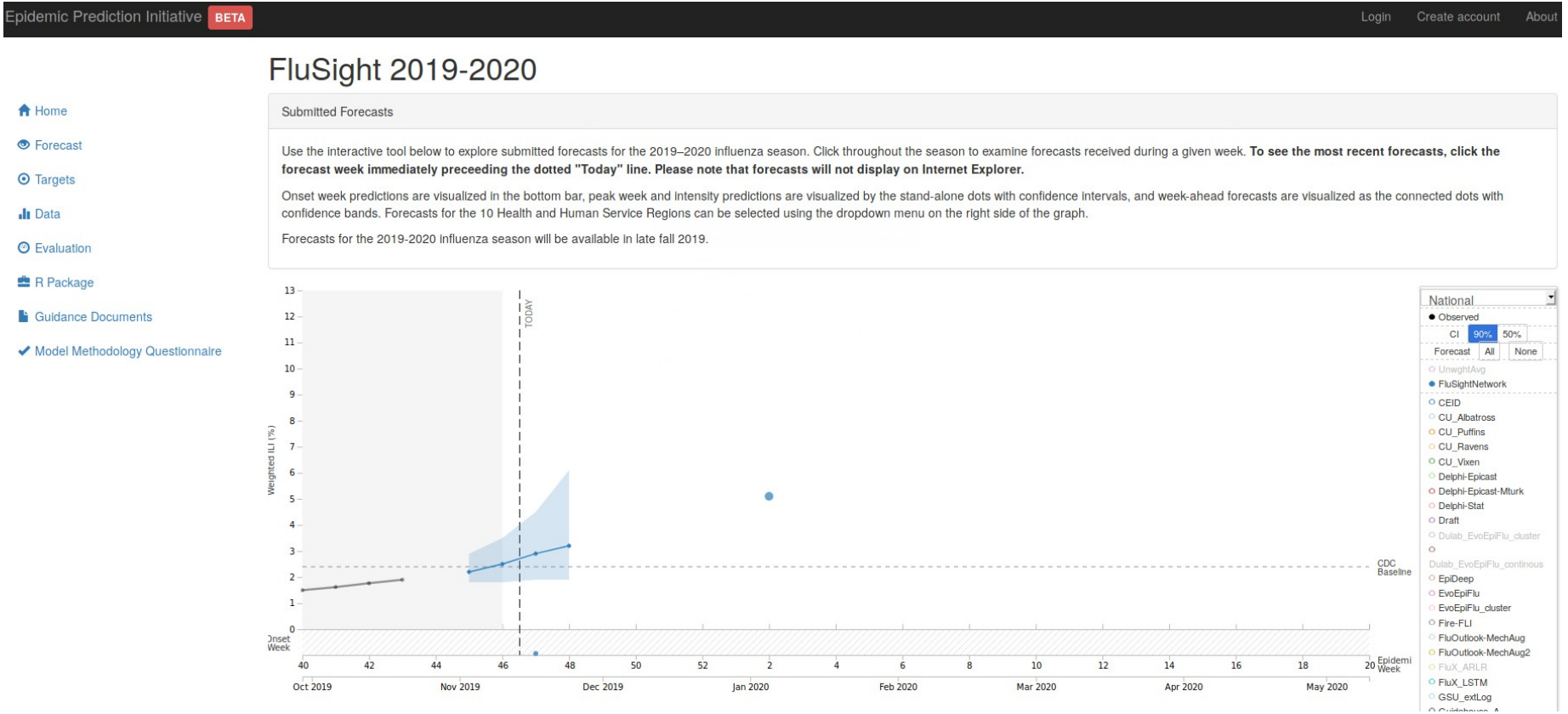
Collaborative Hubs and ensemble predictions

- Example: ECDC Forecasting Hub



Collaborative Hubs and ensemble predictions

- Example: FluSight Influenza forecasting in the US (since 2013)



Collaborative Hubs and ensemble predictions

Collaborative Hubs: Making the Most of Predictive Epidemic Modeling

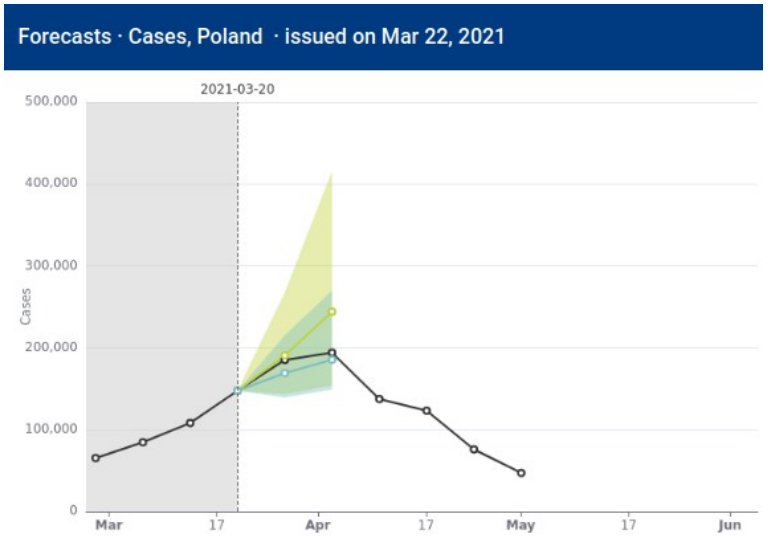
Nicholas G. Reich, PhD, Justin Lessler, Sebastian Funk, PhD, Cecile Viboud, PhD, Alessandro Vespignani, PhD, Ryan J. Tibshirani, PhD, Katriona Shea, PhD, Melanie Schienle, PhD, Michael C. Runge, PhD, Roni Rosenfeld, PhD, Evan L. Ray, PhD, Rene Niehus, PhD, Helen C. Johnson, MRes, MSc, Michael A. Johansson, PhD, Harry Hochheiser, PhD, Lauren Gardner, MSE, PhD, Johannes Bracher, MS, PhD, Rebecca K. Borchering, PhD, and Matthew Biggerstaff, ScD, MPH

American Journal of Public Health, https://doi.org/10.2105/AJPH.2022.306831

What did we learn?

Forecasting emerging epidemics is hard

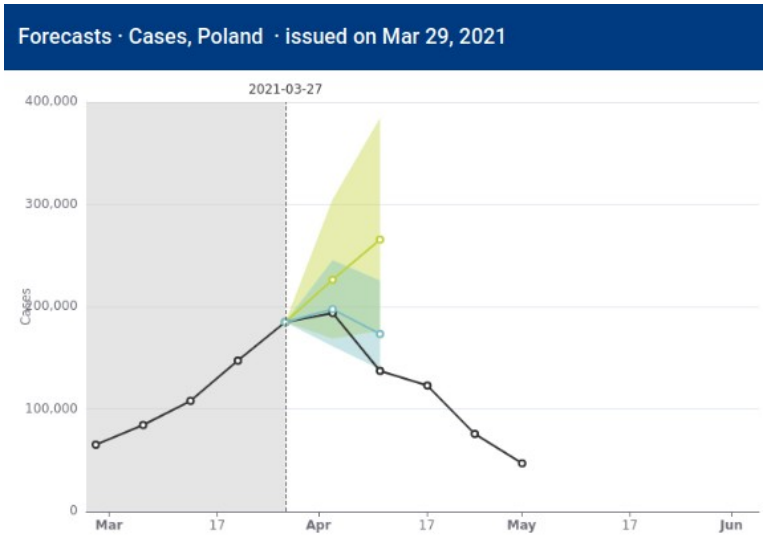
- Nowcasting generally works well.
- Predictions are hard for cases and easier for hospitalizations and deaths.
- Sudden changes are rarely predicted correctly.



Green: ECDC ensemble
Blue: MOCOS

Forecasting emerging epidemics is hard

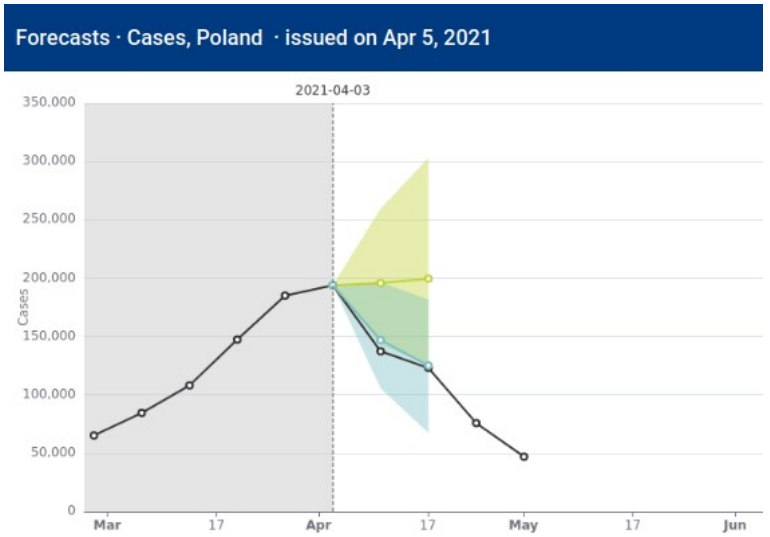
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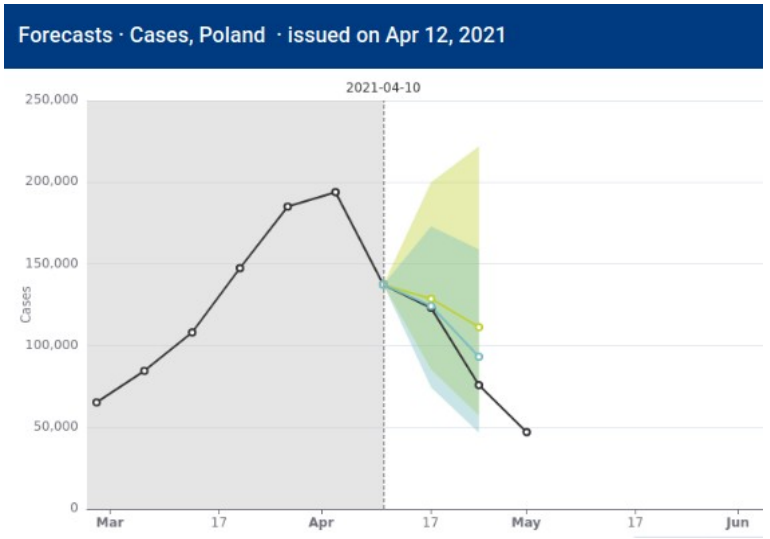
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Why is forecasting epidemics hard?

- The main challenges were known prior to COVID-19:
  - noisy and sparse data.
  - interplay of pathogen, host, environmental and social factors.

The Journal of Infectious Diseases 2016;214(S4):S404–8

Epidemic Forecasting is Messier Than Weather Forecasting: The Role of Human Behavior and Internet Data Streams in Epidemic Forecast

Kelly R. Moran, Geoffrey Fairchild, Nicholas Generous, Kyle Hickmann, Dave Osthus, Reid Priedhorsky, James Hyman, and Sara Y. Del Valle

Ensembles generally work better

- Ensembles overall outperform individual models.
- Their performance is less variable and they quantify uncertainty better.
- But they can’t do miracles.
- Example: nowcasting COVID-19 hospitalizations in Germany.

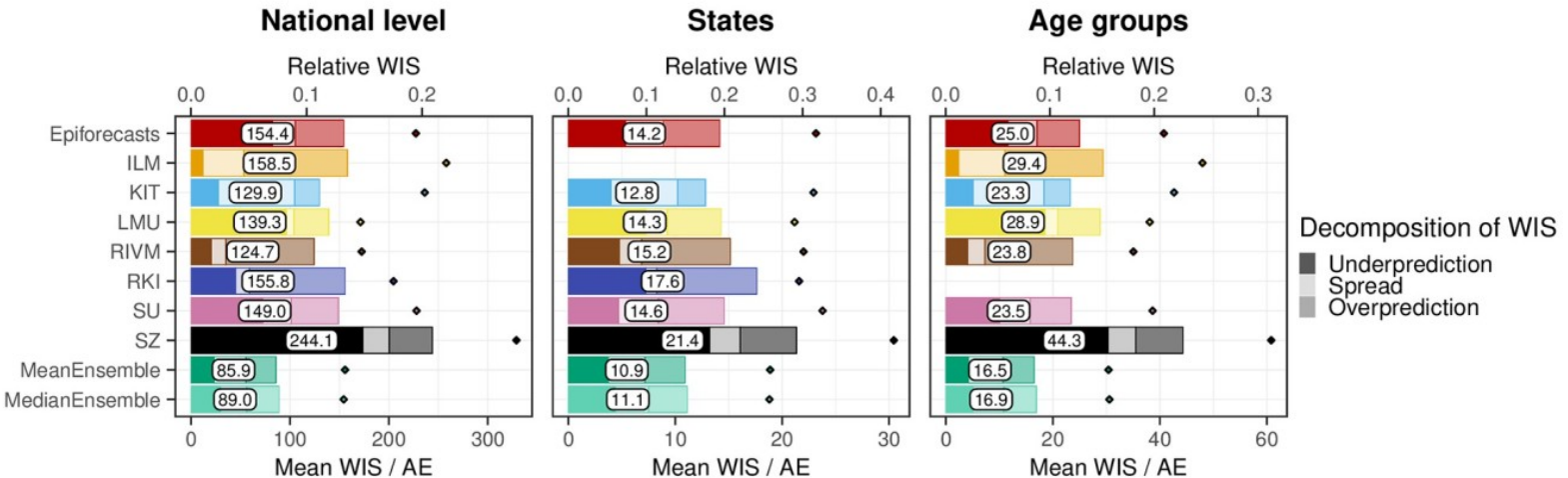


Figure: Wolffram et al (2023), PLOS Computational Biology

What has improved?

What has improved?

- Probabilistic forecasts have replaced point predictions.
- There is less focus on individual models and better comparability as multi-model overviews have become the standard.
- A consensus on aspects like model validation has emerged.
- Research infrastructure and data access have improved.

What is next?

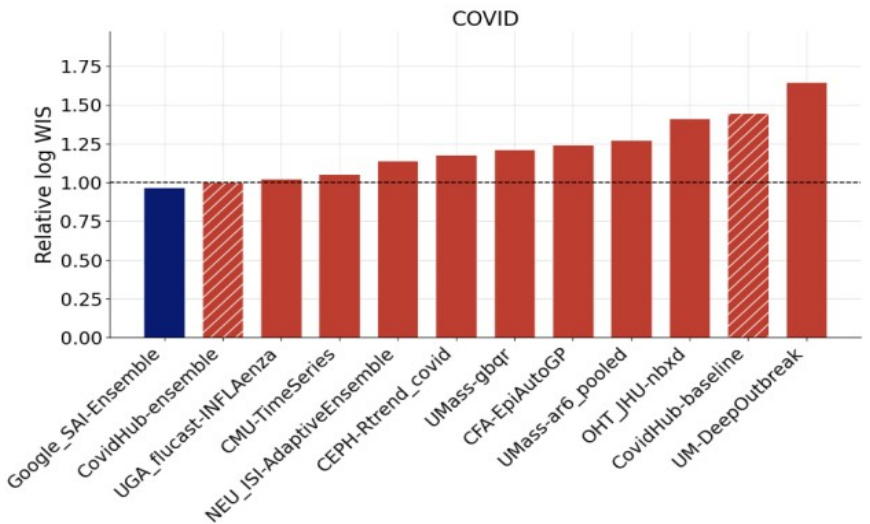
Predicting respiratory diseases after the COVID-19 pandemic

- COVID-19 has become part of the mix of seasonal respiratory diseases.
- Seasonal diseases are **easier** to predict because
  - seasonal patterns are roughly stable.
  - social factors / interventions play less of a role.
- Collaborative forecasting is **harder** to conduct because
  - less data are available.
  - participation is harder to ensure.



AI is catching up (Martinson et al 2026) (This is not my work!)

- AI-generated pipelines can generate forecasts which outperform the ensemble of human-created models.
- But for now, there is no step change in performance.
- My take: improvements will come from new data, not more complex models.



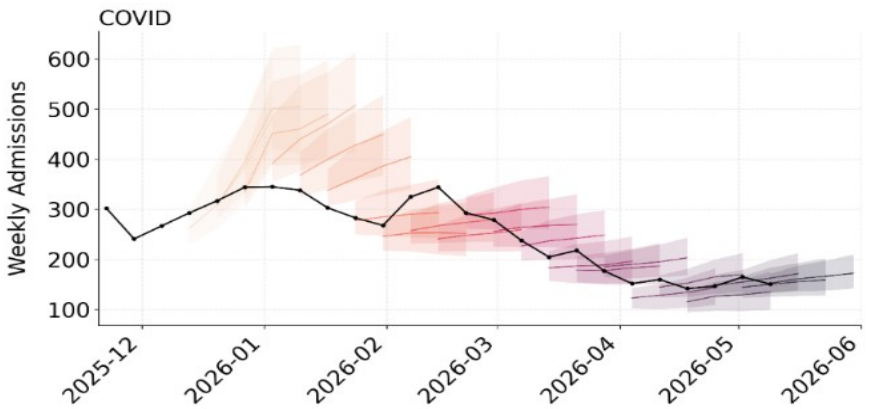


Figure: Martinson et al (2026), arxiv:2605.16238, under CC BY-NC-SA 4.0 License, https://creativecommons.org/licenses/by-nc-sa/4.0/deed.en

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Karlsruhe Institute of Technology

9 December 2026

€ Participation free of charge

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South-West German Infectious Disease Modelling Workshop

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